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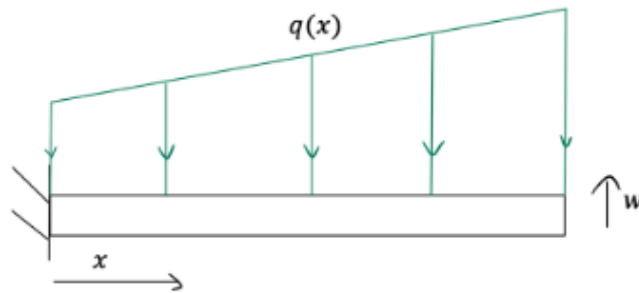
Intro to Finite Element Method (CIV E – 665)

Assignment No. 3



1 Question 1

The transverse deflection $w(x)$ of a cantilever beam due to a distributed load $q(x)$ can be modelled using EQ 1 below. The weak form of EQ 1 is shown in EQ 2 where $v(x)$ represents the weight function. EQ 3 and EQ 4 show a possible approximate solution $W(x)$ and weight function using the Ritz/Galerkin method.



Given equations:

$$EI \frac{d^4 w}{dx^4} = q(x) \quad \text{for } 0 \leq x \leq L \quad (1)$$

Where: $q(x) = -950x - 50$, $L = 1 \text{ m}$ and $EI = 10^6 \text{ Nm}^2$

$$EI \int_0^L \frac{d^2 v}{dx^2} \frac{d^2 w}{dx^2} dx - \left[\frac{dv}{dx} EI \frac{d^2 w}{dx^2} \right]_0^L + \left[v EI \frac{d^3 w}{dx^3} \right]_0^L = \int_0^L v q dx \quad (2)$$

$$W(x) = c_2 x^3 + c_1 x^2 \quad (3)$$

$$v(x) = x^3 + x^2 \quad (4)$$

$$\theta_1 = x^2 \quad \text{and} \quad \theta_2 = x^3$$

1.1 i Show that the approximate solution satisfies the essential boundary conditions at the fixed end ($x = 0$).

For this cantilever beam problem, we at least need to satisfy two essential boundary conditions at the fixed end ($x=0$) to ensure the displacement and slope are both zero.

So, we can get the essential boundary conditions at the fixed: (1) at $x = 0, W(0) = 0$; (2) at $x = 0, W'(0) = 0$

Substitute $x=0$ into $W(x)$

$$W(0) = c_2 \times 0 + c_1 \times 0 = 0$$

So, the essential boundary condition (1) is satisfied.

For essential boundary condition 2, the first derivative of $w(x)$:

$$W'(x) = \frac{dW}{dx} = 3c_2x^2 + 2c_1x \quad (5)$$

Substitute $x=0$ into $W'(x)$

$$W'(x) = \frac{dW}{dx} = 3c_2(0)^2 + 2c_1(0) = 0$$

Therefore, the essential boundary condition 2 is also satisfied.

1.2 ii Solve for the coefficients c_1 and c_2

Based on solution (i), at $x = 0$, we have:

$$\begin{aligned} \left. \frac{d^2w}{dx} \right|_{x=0} &= 0 \\ \left. \frac{d^3w}{dx} \right|_{x=0} &= 0 \\ \left. \frac{d^2v}{dx} \right|_{x=0} &= 0 \\ \left. \frac{d^3v}{dx} \right|_{x=0} &= 0 \end{aligned} \quad (6)$$

applying the essential boundary condition, the Equation 2 can be simplified as:

$$EI \left[\left. \frac{dv}{dx} \frac{d^2w}{dx} \right|_0^L - \int_0^L \frac{dv}{dx} \frac{d^3w}{dx} dx \right] - \left(\left. \frac{dv}{dx} EI \frac{d^2w}{dx^2} \right|_0^L + \left(vEI \frac{d^3w}{dx^3} \right)_0^L \right] = \int_0^L vq dx \quad (7)$$

Using integration by parts on the term $\int_0^L \frac{dv}{dx} \frac{d^3w}{dx} dx$ in Equation 7,

$$\int_0^L \frac{dv}{dx} \frac{d^3w}{dx} dx = \left(v \frac{d^3w}{dx^3} \right)_0^L - \int_0^L \frac{d^2w}{dx^2} \frac{d^2v}{dx^2} dx \quad (8)$$

Substitute Equation 8 to Equation 7, we can get:

$$EI \int_0^L \frac{d^2w}{dx^2} \frac{d^2v}{dx^2} dx = \int_0^L vq dx \quad (9)$$

And then, Building stiffness matrix

$$K_{ij} = EI \int_0^L \frac{d^2 v_i}{dx^2} \frac{d^2 v_j}{dx^2} dx \quad (10)$$

$$F_i = \int_0^L v_i q(x) dx \quad (11)$$

Given $\theta_1 = x^2$ and $\theta_2 = x^3$,

$$\begin{aligned} \frac{d^2 \theta_1}{dx^2} &= 2 \\ \frac{d^2 \theta_2}{dx^2} &= 6x \end{aligned}$$

For each element of the stiffness matrix:

$$\begin{aligned} K_{11} &= EI \int_0^L (2)(2) dx = 4x \Big|_0^{L=1} = 4EI \\ K_{12} &= EI \int_0^L (2)(6x) dx = 6x^2 \Big|_0^{L=1} = 6EI \\ K_{21} &= EI \int_0^L (6x)(2) dx = 6x^2 \Big|_0^{L=1} = 6EI \\ K_{22} &= EI \int_0^L (6x)(6x) dx = 12x^3 \Big|_0^{L=1} = 12EI \end{aligned}$$

so,

$$[K] = EI \begin{bmatrix} 4 & 6 \\ 6 & 12 \end{bmatrix}$$

For $F(x)$,

$$\begin{aligned} F_1 &= \int_0^1 x^2(-950x - 50) dx = \int_0^1 (-950x^3 - 50x^2) dx \\ &= \left[\frac{-950x^4}{4} - \frac{50x^3}{3} \right]_0^1 = -\frac{1525}{6} \\ F_2 &= \int_0^1 x^3(-950x - 50) dx = \int_0^1 (-950x^4 - 50x^3) dx \\ &= \left[\frac{-950x^5}{5} - \frac{50x^4}{4} \right]_0^1 = -\frac{405}{2} \end{aligned}$$

Therefore, the system is:

$$10^6 \begin{bmatrix} 4 & 6 \\ 6 & 12 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -\frac{1525}{6} \\ -\frac{405}{2} \end{bmatrix} \quad (12)$$

Solve the system, $c_1 = -1.52916667 \times 10^{-4}$ and $c_2 = 5.95833333 \times 10^{-5}$

1.3 iii Compare the approximate solution $W(x)$ to the exact solution $w(x)$ obtained in assignment 1 by plotting $w(x)$ and $W(x)$ on the same graph.

Matplotlib is employed to visualize and compare the exact solution $w(x)$ and the approximate solution $W(x)$ derived from the finite element method, as shown in Figure 1. The source code is shown Section 3.1.

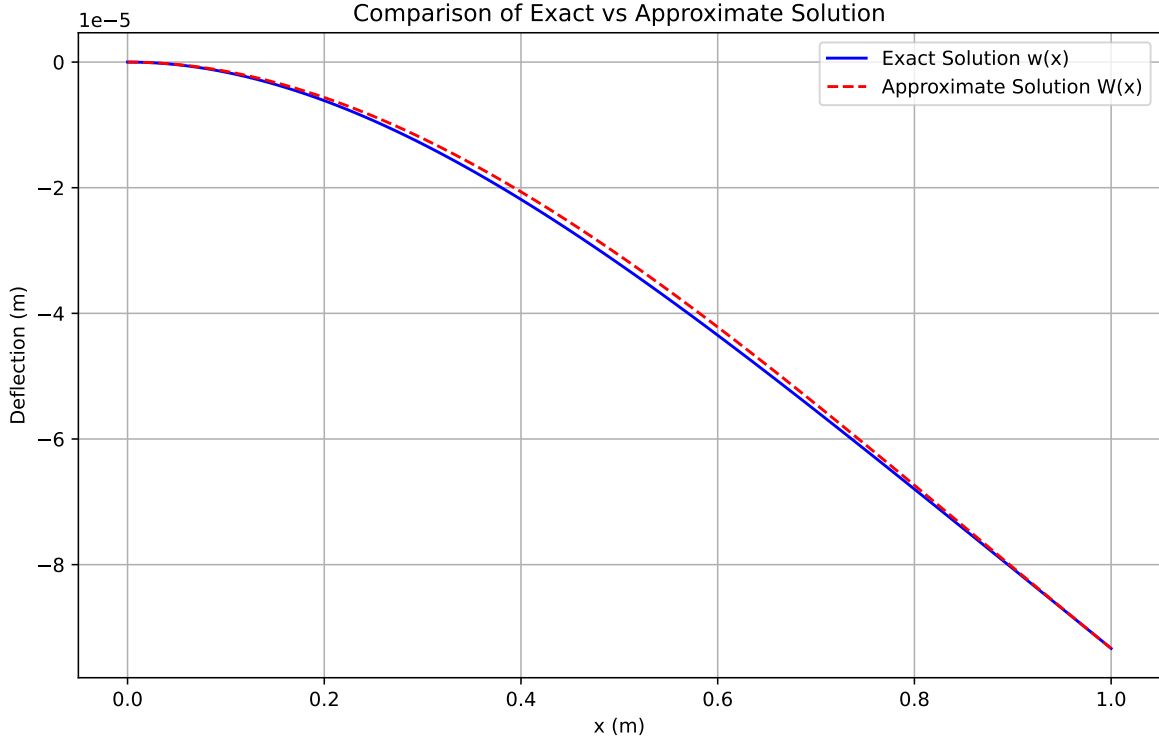


Figure 1: Comparison of Exact vs Approximate Solution

The plot demonstrates that the approximate solution $W(x)$ (red dashed line) is in excellent agreement with the exact solution $w(x)$ (blue solid line).

Both solutions satisfy the essential boundary conditions at the fixed support ($x = 0$), where the deflection is strictly zero. Despite $W(x)$ being a lower-order polynomial (cubic) compared to the exact solution (quintic), it captures the maximum deflection at the free end ($x = 1$) with high accuracy.

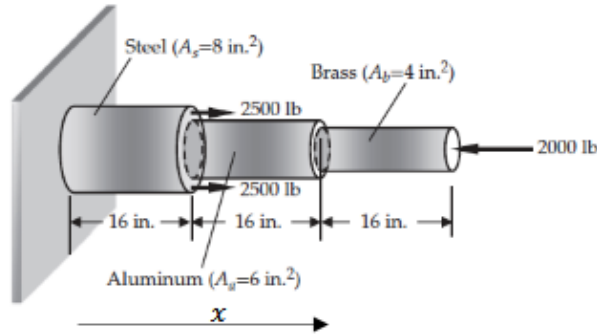
2 Question 2

In this assignment, you can directly use the following formulas to define the stiffness matrix and force vectors for axial bars. There is no need to derive them from the governing equation.

Linear Element	
$[k_{ij}^e] = EA \int_{x_1^e}^{x_2^e} \frac{d\psi_i^e}{dx} \frac{d\psi_j^e}{dx} dx$	$\{f_i\} = \int_{x_1^e}^{x_2^e} \psi_i^e b(x) dx$
$\psi_1^e = \frac{x_2^e - x}{L_e}$	$\psi_2^e = \frac{x - x_1^e}{L_e}$

Question 2 (50 Points)

For the stepped bar shown below, use **3 linear 1D elements** to model the axial displacement in the bar. You can assume that there are no body forces acting on the system.



Since their diameters are different, we need to divide the stepped bar into three elements:

1. Element 1 (steel): from $x = 0$ to $x = 16$ inches, $A_1 = 8 \text{ in}^2$, $E_1 = 30 \times 10^6 \text{ psi}$
2. Element 2 (aluminum): from $x = 16$ inches to $x = 32$ inches, $A_2 = 6 \text{ in}^2$, $E_2 = 10 \times 10^6 \text{ psi}$
3. Element 3 (brass): from $x = 32$ inches to $x = 48$ inches, $A_3 = 4 \text{ in}^2$, $E_3 = 15 \times 10^6 \text{ psi}$

Correspondingly, we have four nodes:

1. Node 1: at $x = 0$ inches
2. Node 2: at $x = 16$ inches
3. Node 3: at $x = 32$ inches
4. Node 4: at $x = 48$ inches

2.1 I Evaluate the element wise stiffness matrices and force vectors

Since these three elements have same length, so for each linear 1D bar element, the stiffness matrix $[k^e]$ is given by:

$$k^e = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

For element 1 (Steel, Nodes 1-2):

$$k^{(1)} = \frac{(30 \times 10^6)(8)}{16} = 15 \times 10^6 \text{ lb/in}$$

$$[k^{(1)}] = 10^6 \begin{bmatrix} 15 & -15 \\ -15 & 15 \end{bmatrix}$$

For element 2 (Aluminum, Nodes 2-3):

$$k^{(2)} = \frac{(10 \times 10^6)(6)}{16} = 3.75 \times 10^6 \text{ lb/in}$$

$$[k^{(2)}] = 10^6 \begin{bmatrix} 3.75 & -3.75 \\ -3.75 & 3.75 \end{bmatrix}$$

For element 3 (Brass, Nodes 3-4):

$$k^{(3)} = \frac{(15 \times 10^6)(4)}{16} = 3.75 \times 10^6 \text{ lb/in}$$

$$[k^{(3)}] = 10^6 \begin{bmatrix} 3.75 & -3.75 \\ -3.75 & 3.75 \end{bmatrix}$$

Element Force Vectors:

The problem states there are no body forces ($b(x) = 0$). Therefore, the distributed force vector $\{f\}$ for every element is zero.

$$\{f^{(1)}\} = \{f^{(2)}\} = \{f^{(3)}\} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

2.2 II Assemble the matrices into a global system

We assemble the local matrices into a global 4×4 stiffness matrix $[K]$.

Global Stiffness Matrix $[K]$:

$$[K] = 10^6 \begin{bmatrix} k_{11}^{(1)} & k_{12}^{(1)} & 0 & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} + k_{11}^{(2)} & k_{12}^{(2)} & 0 \\ 0 & k_{21}^{(2)} & k_{22}^{(2)} + k_{11}^{(3)} & k_{12}^{(3)} \\ 0 & 0 & k_{21}^{(3)} & k_{22}^{(3)} \end{bmatrix} \quad (13)$$

Substituting values:

- $K_{22} = 15 + 3.75 = 18.75$
- $K_{33} = 3.75 + 3.75 = 7.5$

Global Force Vector $\{F\}$:

We assign external point loads to the corresponding nodes.

- **Node 1:** Reaction force F_1
- **Node 2:** Two 2500 lb loads pointing right = +5000 lb
- **Node 3:** No external load = 0
- **Node 4:** One 2000 lb load pointing left = -2000 lb

$$\{F\} = \begin{bmatrix} F_1 \\ 5000 \\ 0 \\ -2000 \end{bmatrix}$$

System Equation:

$$10^6 \begin{bmatrix} 15 & -15 & 0 & 0 \\ -15 & 18.75 & -3.75 & 0 \\ 0 & -3.75 & 7.5 & -3.75 \\ 0 & 0 & -3.75 & 3.75 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} F_1 \\ 5000 \\ 0 \\ -2000 \end{bmatrix}$$

2.3 III Apply the boundary conditions and solve for the nodal displacements

Boundary Condition: The bar is fixed at the wall (Node 1), so displacement is zero.

$$u_1 = 0$$

We can remove the first row and first column (the “reduced” system) to solve for the unknown displacements u_2, u_3, u_4 .

$$10^6 \begin{bmatrix} 18.75 & -3.75 & 0 \\ -3.75 & 7.5 & -3.75 \\ 0 & -3.75 & 3.75 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \\ u_4 \end{bmatrix} = \begin{bmatrix} 5000 \\ 0 \\ -2000 \end{bmatrix}$$

Use Matlab solving (the code please see Section 3.2):

$$u_2 = 2.000000 \times 10^{-04} \quad u_3 = -\frac{10}{3} \times 10^{-04} \quad u_4 = -\frac{26}{3} \times 10^{-04}$$

To construct the full solution

$$U^e(x) = \sum_{i=1}^2 u_i^e \psi_i^e = u_1 \psi_1^e + u_2 \psi_2^e$$

for $u^1(x)$,

$$\psi_1^1 = \frac{x_2^1 - x}{L_1} \quad \text{and} \quad \psi_2^1 = \frac{x - x_1^1}{L_1}$$

We can get:

$$\begin{aligned} u^1(x) &= u_1 \frac{x_2^1 - x}{L_1} + u_2 \frac{x - x_1^1}{L_1} = u^3(x) \\ &= 0 + 2 \times 10^{-4} \frac{x}{16} = 1.25 \times 10^{-5} x \end{aligned}$$

For $u^2(x)$,

$$\psi_1^2 = \frac{x_3^2 - x}{L_2} \quad \text{and} \quad \psi_2^2 = \frac{x - x_2^2}{L_2}$$

We can get:

$$\begin{aligned}
u^2(x) &= u_2 \frac{x_3^2 - x}{L_2} + u_3 \frac{x - x_2^2}{L_2} = \\
&= (2 \times 10^{-4}) \frac{32 - x}{16} + \left(-\frac{10}{3} \times 10^{-4}\right) \frac{x - 16}{16} \\
&= 0.0007\bar{3} - 0.0000\bar{3}x
\end{aligned}$$

For $u^3(x)$,

$$\psi_1^3 = \frac{x_4^3 - x}{L_3} \quad \text{and} \quad \psi_2^3 = \frac{x - x_3^3}{L_3}$$

We can get:

$$\begin{aligned}
u^3(x) &= u_2 \frac{x_4^3 - x}{L_3} + u_3 \frac{x - x_3^3}{L_3} = \\
&= \left(-\frac{10}{3} \times 10^{-4}\right) \frac{48 - x}{16} + \left(-\frac{26}{3} \times 10^{-4}\right) \frac{x - 32}{16} \\
&= 0.0007\bar{3} - 0.0000\bar{3}x
\end{aligned}$$

Therefore, the $u(x)$ for each element is:

$$u(x) = \begin{cases} u^1(x) = 1.25 \times 10^{-5}x & \text{if } x \in [0, 16] \\ u^2(x) = 0.0007\bar{3} - 0.0000\bar{3}x & \text{if } x \in [16, 32] \\ u^3(x) = 0.0007\bar{3} - 0.0000\bar{3}x & \text{if } x \in [32, 48] \end{cases}$$

2.4 IV Plot the displacement, strain, and stress as a function of the axial position x

The displacement, strain, and stress as a function of the axial position x is shown as Figure 2. The plot code please see Section 3.

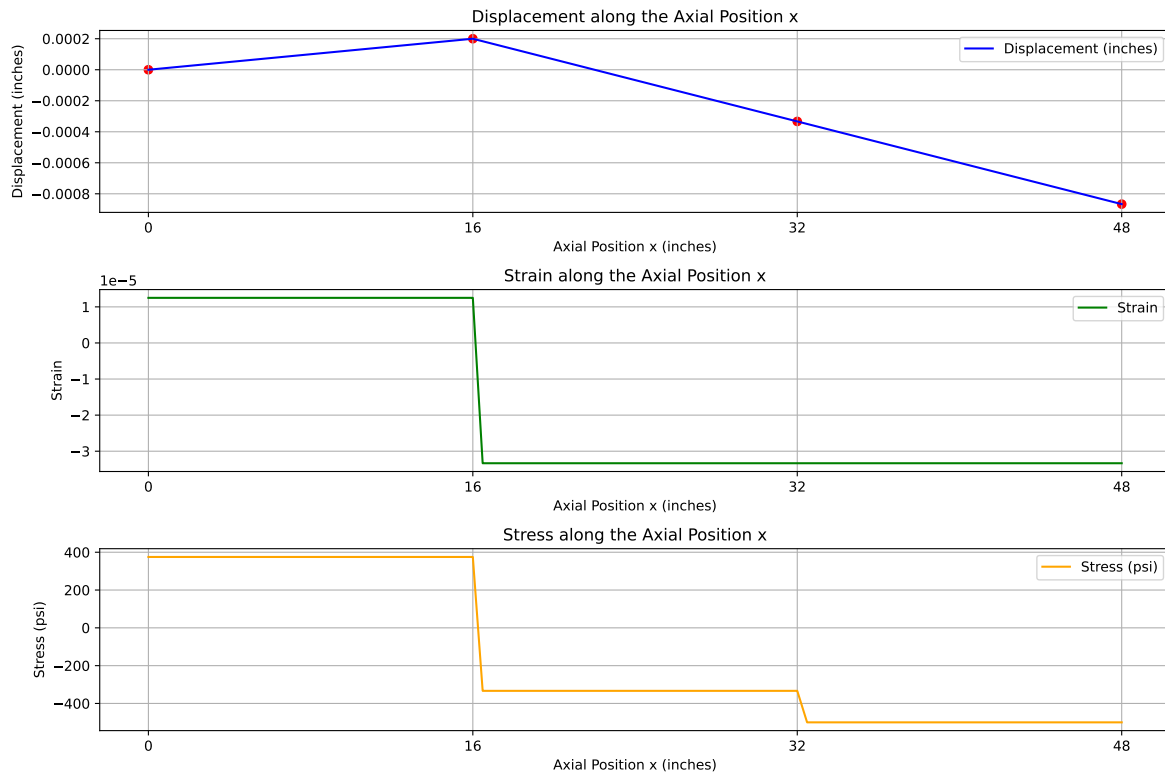


Figure 2: Displacement, strain, and stress along the axial position x

3 Appendix

3.1 The Python code for Question 1 ii

```
#| label: fig-exact-vs-approx
#| caption: Comparison of Exact vs Approximate Solution
#| echo: false
import numpy as np
import matplotlib.pyplot as plt

# Parameters
L = 1
EI = 1e6

# Stiffness Matrix K and Force Vector F
K = EI * np.array([[4, 6], [6, 12]])
F = np.array([-1525/6, -405/2])

# Solve for coefficients c1 and c2
C = np.linalg.solve(K, F)
print( C )
c1, c2 = C
```

```

# Define x range
x = np.linspace(0, L, 100)

# Exact solution w(x)
w_exact = (1/EI) * (-950/120 * x**5 - 50/24 * x**4 + 525/6 * x**3 - 1025/6 * x**2)

# Approximate solution W(x)
W_approx = c1 * x**2 + c2 * x**3

# Plotting
plt.figure(figsize=(10, 6))
plt.plot(x, w_exact, label='Exact Solution w(x)', linestyle='-', color='blue')
plt.plot(x, W_approx, label='Approximate Solution W(x)', linestyle='--', color='red')
plt.title('Comparison of Exact vs Approximate Solution')
plt.xlabel('x (m)')
plt.ylabel('Deflection (m)')
plt.legend()
plt.grid(True)
plt.show()

```

3.2 The Matlab code for Question 2 III

```

% Define the coefficient matrix
A_base = [18.75, -3.75, 0;
          -3.75, 7.5, -3.75;
          0, -3.75, 3.75];
A = 10^6 * A_base;

% Define the constants vector
b = [5000; 0; 2000];

% Solve for u using the backslash operator (Gaussian elimination)
u = A \ b;

% Display the results
fprintf('u2 = %e\n', u(1));
fprintf('u3 = %e\n', u(2));
fprintf('u4 = %e\n', u(3));

```

3.3 The Python code for Question 2 IV

```

#| label: fig-displacement
#| caption: Displacement, strain, and stress along the axial position x

import numpy as np
import matplotlib.pyplot as plt

# Node positions
x_nodes = [0, 16, 32, 48] # in inches
# Displacements at nodes
u_nodes = [0, 2.0e-4, -3.333333e-4, -8.666667e-4] # in inches

```

```

# Create finer x values for plotting
x_fine = np.linspace(0, 48, 100)
# Interpolate displacements
u_fine = np.piecewise(x_fine, [x_fine <= 16, (x_fine > 16) & (x_fine <= 32), x_fine > 32],
                      [lambda x: (u_nodes[1]-u_nodes[0])/(x_nodes[1]-x_nodes[0])*(x - x_nodes[0]) + u_nodes[0],
                        lambda x: (u_nodes[2]-u_nodes[1])/(x_nodes[2]-x_nodes[1])*(x - x_nodes[1]) + u_nodes[1],
                        lambda x: (u_nodes[3]-u_nodes[2])/(x_nodes[3]-x_nodes[2])*(x - x_nodes[2]) + u_nodes[2]])

plt.figure(figsize=(12, 8))
# Displacement plot
plt.subplot(3, 1, 1)
plt.plot(x_fine, u_fine, label='Displacement (inches)', color='blue')
plt.scatter(x_nodes, u_nodes, color='red') # Mark nodes
plt.title('Displacement along the Axial Position x')
plt.xlabel('Axial Position x (inches)')
plt.ylabel('Displacement (inches)')
plt.xticks(x_nodes)
plt.grid()
plt.legend()
# Strain plot
plt.subplot(3, 1, 2)
strain_fine = np.piecewise(x_fine, [x_fine <= 16, (x_fine > 16) & (x_fine <= 32), x_fine > 32],
                           [lambda x: (u_nodes[1]-u_nodes[0])/(x_nodes[1]-x_nodes[0]),
                             lambda x: (u_nodes[2]-u_nodes[1])/(x_nodes[2]-x_nodes[1]),
                             lambda x: (u_nodes[3]-u_nodes[2])/(x_nodes[3]-x_nodes[2])])
plt.plot(x_fine, strain_fine, label='Strain', color='green')
plt.title('Strain along the Axial Position x')
plt.xticks(x_nodes)
plt.xlabel('Axial Position x (inches)')
plt.ylabel('Strain')
plt.grid()
plt.legend()
# Stress plot
plt.subplot(3, 1, 3)
stress_fine = np.piecewise(x_fine, [x_fine <= 16, (x_fine > 16) & (x_fine <= 32), x_fine > 32],
                            [lambda x: 30e6 * (u_nodes[1]-u_nodes[0])/(x_nodes[1]-x_nodes[0]),
                              lambda x: 10e6 * (u_nodes[2]-u_nodes[1])/(x_nodes[2]-x_nodes[1]),
                              lambda x: 15e6 * (u_nodes[3]-u_nodes[2])/(x_nodes[3]-x_nodes[2])])
plt.plot(x_fine, stress_fine, label='Stress (psi)', color='orange')
plt.title('Stress along the Axial Position x')
plt.xlabel('Axial Position x (inches)')
plt.xticks(x_nodes)
plt.ylabel('Stress (psi)')
plt.grid()
plt.legend()
plt.tight_layout()
plt.show()

```